

# Albert J. Lee

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## RESEARCH SUMMARY

ML Ph.D. candidate moving into **robot learning for humanoids**, and quick to become productive in a new field. I joined PIA Robotics, a pre-seed humanoid startup, with no prior robotics experience and within two months built its **full sim-to-real pipeline for humanoid whole-body control** from scratch: RL training in Isaac Lab, sim-to-sim validation in MuJoCo, and deployment on a real humanoid. I also built a text-to-robot motion pipeline that retargets generated human motion through **learned point-cloud correspondence**, trained **BeyondMimic motion-tracking** policies on it, and deployed them on the robot. My Ph.D. work used causal interventions on model internals (activation patching, steering, targeted post-training) to find where specific behaviors come from in LLMs and VLMs; before that, I trained in neuroscience. Seeking research roles in humanoid loco-manipulation and simulation.

## EDUCATION

Ph.D. in Machine Learning, Georgia Institute of Technology

Anticipated 05/2027

B.S. in Neuroscience (Faculty Honors), Georgia Institute of Technology

## RESEARCH EXPERIENCE

Research Intern, PIA Robotics

08/2026–present

*Pre-seed humanoid startup · whole-body control and sim-to-real (Unitree G1)*

- Built the company's **end-to-end sim-to-real pipeline** from scratch: RL training of whole-body control policies in Isaac Lab, sim-to-sim validation in MuJoCo, and deployment on hardware.
- Built a **text-to-robot motion pipeline**: motion generated from text with Kimodo (SOMA-RP-v1) on the SOMA body model, skinned into a human mesh with linear blend skinning, and retargeted to the G1 through **learned point-cloud correspondence** (PointNet autoencoder, following UMR) under contact, joint-limit, and floor constraints, instead of skeleton-based joint mapping.
- Trained **BeyondMimic motion-tracking policies** on the retargeted references and **deployed them on the G1**.
- Whole-body control policies from the sim-to-real pipeline run in the company's **teleoperated hardware demo**, in which the robot walks, squat-walks, and runs while staying balanced.

Research Intern, LG AI Research (Bio Intelligence Lab)

05/2026–08/2026

- Developed **DCCD-J**, a differentiable method for recovering causal graphs from interventional single-cell data under latent confounding and feedback loops, replacing the prior method's alternating covariance estimation, which collapses on real data, with a jointly optimized Cholesky parameterization; confounder-recovery F1 0.93 vs. 0.23. Co-first author, **NeurIPS 2026 Workshop on AI for Drug Discovery (AI4DD)**.
- Replaced expression-based variable selection with **attribution from single-cell foundation models** (scFoundation, scLong) to choose the gene set for regulatory inference.

Graduate Research Assistant (PI: C. S. Mitchell, Ph.D.)

08/2021–present

*Laboratory for Pathology Dynamics, Georgia Tech & Emory University School of Medicine*

- **[Telling Apart What Fine-Tuning Against Wrong Diagnostic Suggestions Gains and Loses]**
  - Showed that fine-tuning Llama-3.1-8B-Instruct (KL-regularized LoRA; also GRPO, and repeated on Qwen2.5-7B-Instruct) to resist wrong diagnostic suggestions raises accuracy under wrong ones (30.5% → 71.5%) but lowers it under correct ones (89.1% → 62.2%), and that what it loses is **the bias toward following suggestions, not discrimination** (AUROC 0.684 → 0.741).
  - Located the two changes at different tokens with **cross-model activation patching**, then **steered only the suggestion tokens** to recover 59% of the loss while keeping 79% of the discrimination gain, also on 951 held-out pairs.
  - *Under review at ACL Rolling Review (2026).*
- **[Goal-Aware LLM Alignment under Evidence and Eligibility Constraints]**
  - *Ongoing research project | 2026–present*
  - Built **counterfactual evaluations** of whether an LLM updates treatment-eligibility judgments when the rules change, but not when only the patient's stated goal changes.
  - Localized the failure to about 2% of attention heads with **activation patching** and post-train only those heads (SFT + RL with verifiable rewards), against budget-matched random-head and full-parameter baselines.

- **[Recovery Is Not Provenance: A Source-Substitution Audit of VLM Activation Patching in Medical VQA]**
  - Showed that activation patching that repairs a VLM's answer need not restore evidence from the image: with **source substitution**, which swaps the donor case while fixing the patch site, different-case donors reproduced 98–103% of same-case recovery across 5 VLMs.
  - *Under review at AAAI 2027.*
- **[Seven Medical Reasoning Biases, One Causal Direction]**
  - Found that one activation direction reduced cue-following across seven medical QA cue types and three LLM families; compared shared, cue-specific, and random directions to **separate what is detectable from what works as an intervention**.
  - *Under review at ACL Rolling Review (2026).*
- **[HiCo-EL: Encoding Ontology Structure into Bi-Encoder Entity Linking]**
  - Hierarchical contrastive pretraining that organizes each encoder space by the ontology's type hierarchy; +3.7% R@1 on MedMentions and **+9.6% R@1 on diseases unseen in training**. Supervised two junior researchers. EMNLP 2026.
- **Earlier work (2021–2025):** contributed to the lab's biomedical text-mining tools (CCS Explorer, TrialSieve, SemNet Explorer), with HiCo-EL as the entity-linking step; interpretable ML for cognitive assessment (first author, J. Clin. Med. 2025).

#### **Computational Neuroscience Rotating Fellow (PI: C. Rozell, Ph.D.)**

01/2021–05/2021

*Department of Electrical Engineering, Georgia Tech*

- Applied embedding-based similarity models and probabilistic latent-variable methods to characterize clinical depression phenotypes.

#### **Undergraduate Research Fellow and ALS Team Lead (PI: C. S. Mitchell, Ph.D.)**

12/2016–05/2020

*Laboratory for Pathology Dynamics, Georgia Tech & Emory University School of Medicine*

- Led ML-driven meta-analysis of heterogeneous preclinical datasets (>2,000 data points) and multi-scale dynamical-systems models of ALS.
- Led an NIH grant and a team of 5 researchers, resulting in 9 publications.

#### **SKILLS**

- **Robotics:** Isaac Lab, MuJoCo; RL for humanoid whole-body control and motion tracking (BeyondMimic); sim-to-sim and sim-to-real transfer on a Unitree G1; text-to-motion generation (Kimodo, SOMA); correspondence-based motion retargeting (linear blend skinning, PointNet autoencoder, UMR).
- **Machine learning:** Python, PyTorch, HuggingFace, Accelerate, DeepSpeed, multi-GPU training, W&B, MATLAB, R; SFT, GRPO (RL with verifiable rewards), LoRA; activation patching and causal interventions; controlled evaluation design.
- **Languages:** Native in Korean; fluent in English.

#### **PUBLICATIONS**

##### **Under review:**

- [1] **Lee, A. J. B.**, & Mitchell, C. S. Recovery Is Not Provenance: A Source-Substitution Audit of VLM Activation Patching in Medical VQA. Under review at AAAI 2027.
- [2] **Lee, A. J. B.**, & Mitchell, C. S. Seven Medical Reasoning Biases, One Causal Direction. Under review at ACL Rolling Review (2026).
- [3] **Lee, A. J. B.**, & Mitchell, C. S. Telling Apart What Fine-Tuning Against Wrong Diagnostic Suggestions Gains and Loses. Under review at ACL Rolling Review (2026).

##### **Peer-reviewed conference papers:**

- [4] **Lee, A. J. B.**, Bathala, P., Malhotra, P., Ye, C., & Mitchell, C. S. (2026). HiCo-EL: Encoding Ontology Structure into Bi-Encoder Entity Linking. EMNLP 2026.
- [5] Al-Hussaini, I., An, D. N., **Lee, A. J.**, Bi, S., & Mitchell, C. S. (2022). CCS Explorer: Relevance Prediction, Extractive Summarization, and Named Entity Recognition from Clinical Cohort Studies. IEEE International Conference on Big Data (IEEE BigData 2022), Oral.

##### **Peer-reviewed workshop papers:**

- [6] **Lee, A. J. B.\***, Kang, J.\*, Kim, S.\*, & Jeong, H. (2026). Separating Co-expression from Regulation in Cyclic Causal Discovery with Latent Confounders. NeurIPS 2026 Workshop on AI for Drug Discovery (AI4DD). \*Equal contribution.

##### **Peer-reviewed journal articles:**

- [7] He, X., Camacho, D., Moukheiber, L., Iyer, M., Zhao, B., Ye, C., Nursal, B., Guo, X., **Lee, A. J. B.**, & Mitchell, C. S. (2026). SemNet Explorer: An Evidence-Grounded Knowledge Graph–LLM Framework for Multi-Scale Mechanistic Reporting Across Biomedical Domains. Big Data and Cognitive Computing, 10(6), 171.

- [8] Fay, M. E., Tandon, R., Latham, T., **Lee, A. J.**, Rankine-Mullins, A. E., Reid, M., Mitchell, C. S., Ware, R. E., & Lam, W. A. (2026). Artificial Intelligence-Based Analysis of Central Nervous System Vasculopathy in Pediatric Sickle Cell Anemia. *American Journal of Hematology*, 101(5), 998–1004.
- [9] **Lee, A. J. B.**, Zhao, B., Lah, J. J., John, S. E., Loring, D. W., & Mitchell, C. S. (2025). Education, Sex, and Age Shape Rey Complex Figure Performance in Cognitively Normal Adults: An Interpretable Machine Learning Study. *Journal of Clinical Medicine*, 14(21), 7562.
- [10] Kartchner, D., Turner, H., Ye, C., Al-Hussaini, I., Nursal, B., **Lee, A. J. B.**, Deng, J., Curtis, C., Cho, H., Duvaris, E. L., Jackson, C., Shanks, C. E., Tan, S. Y., Ramalingam, S., & Mitchell, C. S. (2025). TrialSieve: A Comprehensive Biomedical Information Extraction Framework for PICO, Meta-Analysis, and Drug Repurposing. *Bioengineering*, 12(5), 486.
- [11] **Lee, A. J. B.**, Bi, S., Ridgeway, E., Al-Hussaini, I., Deshpande, S., Krueger, A., Khatri, A., Tsui, D., Deng, J., & Mitchell, C. S. (2025). Restoring Homeostasis: Treating Amyotrophic Lateral Sclerosis by Resolving Dynamic Regulatory Instability. *International Journal of Molecular Sciences*, 26(3), 872.
- [12] **Lee, A. J. B.**, Kittel, T. E., Kim, R. B., Bach, T. N., Zhang, T., & Mitchell, C. S. (2022). Comparing Therapeutic Modulators of the SOD1 G93A Amyotrophic Lateral Sclerosis Mouse Pathophysiology. *Frontiers in Neuroscience*, 16, 1111763.

#### **Conference Proceedings with Presentation:**

- [1] Deng, J., Bi, S., Tsui, D., **Lee, A.**, Al-Hussaini, I., & Mitchell, C. S. (2022). Homeostatic re-stabilization of SOD1-G93A ALS via combination therapy with repurposed drugs. *International Motoneuron Society*, Banff, Canada.
- [2] Krueger, A., Bi, S., **Lee, A.**, & Mitchell, C. S. (2021). In silico multi-factorial, multi-scalar models of system stability for ALS combination therapies. *AAPM Conference on Data Science in Medical Physics*.
- [3] Bi, S., Krueger, A., **Lee, A.**, & Mitchell, C. S. (2021). In-silico multi-factorial system stability to identify promising SOD1-G93A ALS mouse combination treatments. *Society for Neuroscience*.
- [4] Bi, S., Krueger, A., Ridgeway, E., Khatri, A., Deshpande, S., **Lee, A.**, & Mitchell, C. S. (2021). Dynamic Network Stability Analysis for Prioritizing Experimental Combination Therapies for ALS. *American Neurological Association*, 146th Annual Meeting.
- [5] **Lee, A.**, Lee, B., Ridgeway, E., & Mitchell, C. S. (2019). Machine learning enhanced dynamic meta-analysis for preclinical ALS combination therapy optimization. *Society for Neuroscience*, Chicago, IL.
- [6] **Lee, A.**, Lee, B., Ridgeway, E., & Mitchell, C. S. (2019). Dynamic Meta-Analysis to Assess Multi-factorial Neuropathology. *American Neurological Association*, St. Louis, MO.
- [7] **Lee, A.**, Lee, B., Ridgeway, E., & Mitchell, C. S. (2019). Machine learning enhanced dynamic meta-analysis for multi-factorial disease prediction. *MLSE Conference*, Atlanta, GA.
- [8] Deshpande, S., Lee, B., **Lee, A.**, Ridgeway, E., & Mitchell, C. S. (2019). Dynamic Meta-Analysis to Assess Multi-factorial Disease. *SEMDA Medtech Conference*, Alpharetta, GA.
- [9] **Lee, A.**, & Mitchell, C. S. (2019). Discovering potential combinational treatments of ALS using a computational model of G93A mouse. *Georgia Tech Undergraduate Research Spring Symposium*, Atlanta, GA.
- [10] **Lee, A.**, Travaglino, S., & Mitchell, C. S. (2018). Dynamic Meta-Analysis Approach to In Silico Models of Temporal Multi-scalar Neurophysiology. *Biomedical Engineering Society*, Atlanta, GA.
- [11] **Lee, A.**, Travaglino, S., & Mitchell, C. S. (2017). Towards syncing polytherapy to the multi-factorial dynamics of the SOD1 G93A ALS pathology. *Society for Neuroscience*, Washington, D.C.
- [12] Kittel, T., **Lee, A.**, Nguyen-Bach, T., & Mitchell, C. S. (2017). Data enabled design of multi-factorial disease treatment paradigms: a case study of ALS. *National Academy of Engineering Regional Meeting*, Atlanta, GA.

#### **TEACHING EXPERIENCE**

##### **Graduate Teaching Assistant, Georgia Tech, Dept. of Biomedical Engineering**

Summer 2025

- Graded homework, held weekly office hours, and reviewed concepts for BMED2400 (Bioengineering Statistics).
- Nominated for departmental award recognizing outstanding teaching and mentorship.
- Supervised two junior researchers on HiCo-EL through to an EMNLP 2026 acceptance.

#### **HONORS / FELLOWSHIPS**

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|------------------------------------------------------------------------------|---------|
| BME Department Outstanding Teaching & Mentorship Award — Nominee             | 2025    |
| National Science Foundation Graduate Research Fellowship — Honorable Mention | 05/2020 |
| NIH T32 Training Fellowship in Computational Neuroengineering                | 05/2020 |
| President's Undergraduate Research Project Award                             |         |
| President's Undergraduate Research Travel Award                              | 10/2017 |
| Zell Miller Scholarship (full-tuition merit-based scholarship)               |         |